

Biology Ch. 7: Membrane Transport

7.1 The plasma membrane and selective permeability

Every cell is enclosed by a plasma membrane made of a phospholipid bilayer studded with proteins. The membrane is selectively permeable, meaning it allows some substances to cross freely while blocking or regulating others. Small nonpolar molecules such as oxygen and carbon dioxide slip through the lipid bilayer easily, while ions and large polar molecules like glucose cannot cross on their own.

This selectivity is what makes the cell's interior chemically different from its surroundings, and that difference is essential for everything the cell does, from generating energy to sending signals. Because the membrane restricts free movement, cells rely on several distinct transport strategies to move materials in and out, each suited to a different kind of cargo.

7.2 Diffusion and osmosis

Diffusion is the net movement of molecules from an area of higher concentration to an area of lower concentration, driven by random molecular motion and requiring no input of cellular energy. Over time, diffusion tends to spread a substance out until its concentration is equal everywhere, a state called equilibrium.

Osmosis is the diffusion of water across a selectively permeable membrane, from a region of lower solute concentration to a region of higher solute concentration. Water moves to balance solute concentration on both sides of the membrane, even though the solutes themselves may not be able to cross. A cell placed in a hypertonic solution loses water and shrinks, while a cell placed in a hypotonic solution gains water and may swell or burst.

7.2 Diffusion and osmosis, continued: rates and equilibrium

The rate of diffusion depends on the steepness of the concentration gradient, the temperature, and the size of the diffusing molecules: steeper gradients, higher temperatures, and smaller molecules all diffuse faster. Diffusion never truly stops once equilibrium is reached — molecules keep moving — but with no net concentration difference left, there is no further net movement in either direction.

Osmosis follows the same logic applied specifically to water. If a cell's cytoplasm and its surroundings have equal solute concentration (an isotonic solution), water moves into and out of the cell at equal rates, and the cell's volume stays stable. Cells and organisms regulate their internal solute concentration carefully because uncontrolled osmosis can be damaging or fatal.

7.3 Facilitated diffusion versus active transport

Facilitated diffusion moves substances across the membrane through specific channel or carrier proteins, still moving down the concentration gradient and still requiring no direct energy input from the cell. Glucose, for instance, is too large and too polar to cross the lipid bilayer directly, so it relies on carrier proteins called glucose transporters to move into the cell whenever the outside concentration is higher.

Active transport moves substances against their concentration gradient, from low concentration to high concentration, and this uphill movement requires energy, usually supplied directly by ATP. The sodium-potassium pump is a classic example: it uses ATP to pump sodium ions out of the cell and potassium ions in, both against their gradients, maintaining the concentration differences that neurons depend on to fire.

The key distinction between facilitated diffusion and active transport is not which molecules move, but which direction they move relative to their gradient and whether the cell must spend energy to make it happen. A useful shortcut: if transport requires ATP or another energy source, it is active; if it simply provides an easier path down an existing gradient, it is facilitated diffusion.

7.4 Endocytosis and exocytosis

Some cargo is too large to pass through any membrane protein, so cells move it in bulk using vesicles instead. Endocytosis is the process by which the cell membrane folds inward to engulf material from outside the cell, pinching off to form a vesicle that carries the material into the cytoplasm. This is how cells take in large particles, other cells, or big clusters of molecules.

Exocytosis is the reverse process: a vesicle inside the cell fuses with the plasma membrane and releases its contents to the outside. Exocytosis is how cells secrete hormones, neurotransmitters, and waste products, and how proteins built inside the cell reach the cell surface or leave it entirely.

Both endocytosis and exocytosis require energy and directly reshape the membrane itself, which sets them apart from diffusion, osmosis, and even active transport through pumps: those three move individual molecules or ions through the membrane, while endocytosis and exocytosis move whole vesicles across it.